SHADOW REMOVAL USING PCA

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```
clc; clear; close all;



inputFolder  = 'C:\Users\Divya Rangumudri\Downloads\project_shadow_removal\shadow_images\input';
outputFolder = 'C:\Users\Divya Rangumudri\Downloads\project_shadow_removal\shadow_images\output';



if ~exist(outputFolder,'dir'), mkdir(outputFolder); end


% Load images
files = [dir(fullfile(inputFolder,'*.jpg')); ...
        dir(fullfile(inputFolder,'*.jpeg')); ...
        dir(fullfile(inputFolder,'*.png'))];


if isempty(files)
   error(' No images found. Put .jpg/.jpeg/.png into: %s', inputFolder);
end


fprintf('Found %d image(s).\n', numel(files));


for k = 1:numel(files)


   fname = files(k).name;
   fprintf("\nProcessing: %s\n", fname);
```

```
I = im2double(imread(fullfile(inputFolder, fname)));
if size(I,3)==1, I = repmat(I,[1 1 3]); end

% extract rgb channels
R = I(:,:,1) + eps;  G = I(:,:,2) + eps;  B = I(:,:,3) + eps;

C1 = log(R) - log(G); % produces shadow reduced features
C2 = log(B) - log(G);

[h,w] = size(C1);
X = [C1(:) C2(:)];
Xc = X - mean(X,1);% subtract mean

C = (Xc.'*Xc)/(numel(Xc)/2-1);%covariance to find eigen vectors to give directions
[V,~] = eig(C);
e2 = V(:,1); %illumination invariant direction

invImg = reshape(Xc*e2, h, w);% project image onto pca axis
invImg = invImg - min(invImg(:));
invImg = invImg / (max(invImg(:))+eps);%makes image displayable


grayOrig = max(I,[],3) + 1e-6;%extract brightness
grayInv  = invImg + 1e-6;

ratio = grayInv ./ grayOrig;% brighten shadow area

ratioSmooth = imbilatfilt(ratio, 0.28, 24);%smooths without blurring

ratioSmooth = max(min(ratioSmooth, 1.75), 0.70);%avoids extreme brightening
```

```
    finalImg = I .* ratioSmooth;
    finalImg = max(min(finalImg,1),0);
    figure('Name',['Shadow Removal Result: ' fname], 'NumberTitle','off');
    subplot(1,2,1); imshow(I); title('Original');
    subplot(1,2,2); imshow(finalImg); title('Shadow Removed (Improved)');
    [~,stem,~] = fileparts(fname);
    imwrite(finalImg, fullfile(outputFolder, [stem '_deshadow_final.png']));
    fprintf("Saved: %s_deshadow_final.png\n", stem);

end

fprintf("\nDone! Results shown in MATLAB and saved here:\n%s\n", outputFolder);
```

OUTPUT:

Saved: image1_deshadow_final.png

Processing: myface1.jpg.jpg

Original | Shadow Removed (Improved)

Saved: myface1.jpg_deshadow_final.png

Processing: myface2.jpg

Original | Shadow Removed (Improved)